UNIT 2. IOT AND M2M

2.1 INTRODUCTION M2M

M2M offers point to point exchanges of information without internet connection. It also does not require any sort of manual human intervention. M2M communication has been used in telemetry for a long time. M2M was found before the internet of things and hence one could say that IoT improves the functionality of M2M by offering more complicated functions. Some of the applications of IoT are in traffic control and smart agriculture.

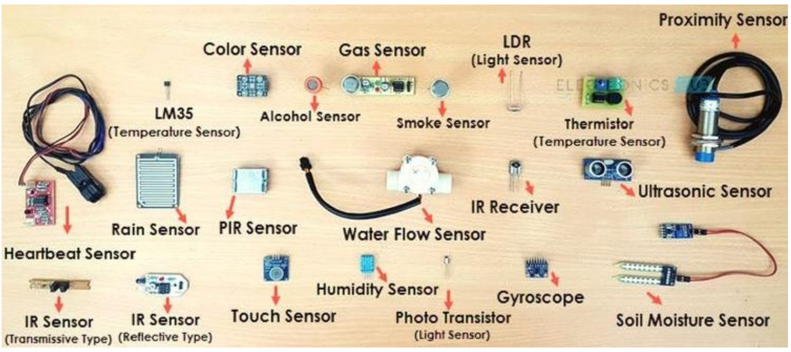
WORKING OF M2M

- M2M devices send data across a network by sensing information. In order to send the data the machines use public networks such as cellular and ethernet.
- M2M comprises components such as RFID, sensors, Wi-Fi communication links and automated computer software programs that translate the incoming data to generate responses or actions.
- Telemetry is one of the most renowned M2M communication. It has been in use since the beginning of the last century to transmit data. Developers used telephone lines for communication and later moved to radio wave transmission signals in order to monitor the performance of the data that is gathered from remote locations.
- The arrival of the internet improved the standards of wireless technology and now wireless communication is used in everyday real life applications such as hospitals, cities, stations, roads and so on.

APPLICATIONS AND EXAMPLES OF M2M

- The most common use of M2M is remote monitoring. For example, a vending machine can notify the merchant in case a product is out of stock. M2M is also used in supply chain management and warehouse management systems.
- Companies and organizations depend on M2M machines as they use less energy such as oil and gas. M2M machines can also bill users such as smart metres. M2M machines can be used in factories to detect conditions such as pressure, temperature, and equipment status.
- In telemedicine, M2M allows remote check up on patients. It allows dispensing medicine and allows doctors to track the health status of patients.
- M2M also contributes heavily to financial activities to allow different purchasing options and examples include Google Wallet and Apple Pay.
- Smart home systems incorporate M2M communication. It allows devices and home appliances to communicate with each and send message over a network
- Finally, M2M also plays a huge part in robotics, traffic management, remote-control software, logistics, fleet management, and automation.

2.2 INTRODUCTION TO SENSOR TECHNOLOGY



- a sensor is a device that detects or measures a physical property and records, indicates, or responds to it by transmitting a resulting impulse. It converts a physical specification (such as temperature, speed, or distance) into a signal, which can be electrically recorded.
- Sensors have been around for decades, but recent IoT solutions have boosted their popularity and the need for more sophisticated and technologically-loaded sensors. Sensors today must encompass the ability to communicate with other sensors and platforms within an entire 'sensor ecosystem'. The role of the sensor is extremely important in the overall IoT picture.
- The number and variety of sensors that gather information to be measured and analyzed is growing significantly, especially with the rise of automation. For this reason, most types of sensors fall under a sensor classification system.

CLASSIFICATION OF SENSORS

Sensors can be ranked into various classification systems, but for the sake of simplification, we have divided them into 5 core classifications.

1. **Active and Passive Sensors**

- Active Sensors (also known as parametric sensors) are sensors that require an external power source to operate. Examples of active sensors include GPS sensors and radar sensors.
- Passive Sensors (also called self-generated sensors) generate their own electric signal and do not require any external power source. Examples of passive sensors include thermal sensors, electric field sensing, and metal detecting.

2. **Contact and Non-Contact Sensors**

- Contact Sensors are those that require physical contact with their stimulus. Familiar examples of contact sensors are temperature and strain gauge sensors.
- Non-Contact Sensors, on the other hand, require no physical contact. These types of sensors include optical and magnetic sensors, as well as infrared thermometers.

3. **Absolute and Relative Sensors**

- Absolute Sensors mimic its name by providing an absolute reading of its stimulus. For example, a thermistor always measures the exact, or absolute, temperature reading.
- Relative Sensors provide measurement to a fixed or variable measurement. An example of a relative sensor would be a thermocouple, where the temperature difference is measured, not the actual temperature.

4. **Analog and Digital Sensors**

- Analog Sensors produce continuous analog output signals, proportional to its measurement. A few examples of analog sensors are: accelerometers, pressure sensors, light, and sound sensors.
- Digital Sensors (also known as electronic or electrochemical sensors) convert the data transmission, digitally. Examples include digital accelerometers, pressure, and temperature sensors.

5. **Miscellaneous Sensors**

- Of course, there are plenty of other types of sensors in the field and they would fall under the 'other' or miscellaneous category. These include electric, biological, chemical, radioactive and more.

  The most common sensors are typically used for measuring physical properties such as temperature, heat, and conduction. Here are a few examples of these commonly used sensors:

- Temperature Sensor
- CO_2 Sensor
- Proximity Sensor
- Accelerometer
- IR Sensor (Infrared Sensor)
- Pressure Sensor
- Light Sensor
- Smoke, Gas, and Alcohol Sensor
- Touch Sensor

2.3 DIFFERENCE BETWEEN IOT AND M2M

Basis of	IoT	M2M
Abbreviation	Internet of Things	Machine to Machine
Intelligence	Devices have objects that are responsible for decision making	Some degree of intelligence is observed in this.
Connection type used	The connection is via Network and using various communication types.	The connection is a point to point
Communication protocol used	Internet protocols are used such as HTTP, FTP, and Telnet.	Traditional protocols and communication technology techniques are used
Data Sharing	Data is shared between other applications that are used to improve the end-user experience.	Data is shared with only the communicating parties.
Internet	Internet connection is required for communication	Devices are not dependent on the Internet.
Type of Communication	It supports cloud communication	It supports point-to-point communication.

Basis of	IoT	M2M
Computer System	Involves the usage of both Hardware and Software.	Mostly hardware-based technology
Scope	A large number of devices yet scope is large.	Limited Scope for devices.
Business Type used	Business 2 Business(B2B) and Business 2 Consumer(B2C)	Business 2 Business (B2B)
Open API support	Supports Open API integrations.	There is no support for Open APIs
Examples	Smart wearables, Big Data and Cloud, Smart wearables, smart cities, smart cars, smart homes and so on etc.	Sensors, Data and Information, etc.
Communication protocols	TCP/IP protocols such as HTTP, HTTPS, FTP and Telnet	Traditional and less secure communication protocols are used

2.4 SECURITY FOR IOT

1. IoT Network Security

IoT network security is challenging than traditional network security as communication protocols, standards, and device capabilities have a more extensive range, all of which pose significant issues and increased complexity. It involves securing the network connection that connects the IoT devices with the back-end systems on the internet. Capabilities include traditional endpoint security features like antivirus and antimalware as well as firewalls and intrusion prevention and detection systems. Sample vendors are Cisco, Darktrace, and Senrio.

2. IoT authentication

It grants users to authenticate IoT devices, including managing multiple users for a single device, ranging from multiple static passwords to more robust authentication mechanisms like two-factor authentication, digital certificates, and biometrics. Unlike most enterprise networks where authentication processes involve a human being entering a credential, many IoT authentication scenarios ar M2M based and do not involve any human intervention. Sample vendors: Baimos Technologies, Covisint, Entrust Datacard, and Gemalto.

3. IoT Encryption

Encrypting data at rest and transit between IoT edge devices and back-end systems using standard cryptographic algorithms, maintaining data integrity, and preventing data sniffing by hackers. Several IoT devices and hardware profiles limit the ability to have standard encryption processes and protocols. Further, all IoT encryption must be accompanied by equivalent full encryption key lifecycle management processes, since poor key management would reduce overall security. Sample vendors: Cisco, HPE.

4. IoT Security Analytics

This technology involves collecting, aggregating, monitoring, and normalizing data from IoT devices and providing actionable reporting and alerting on suspicious activity or when activity falls outside established policies.

These solutions add sophisticated machine learning, artificial intelligence, and big data techniques providing more predictive modeling and anomaly detection, but such capabilities are still emerging. IoT security analytics would increasingly be required to detect IoT-specific attacks and intrusions that are not identified by traditional network security solutions such as firewalls. Sample vendors: Cisco, Indegy, Kaspersky Lab, SAP, and Senrio.

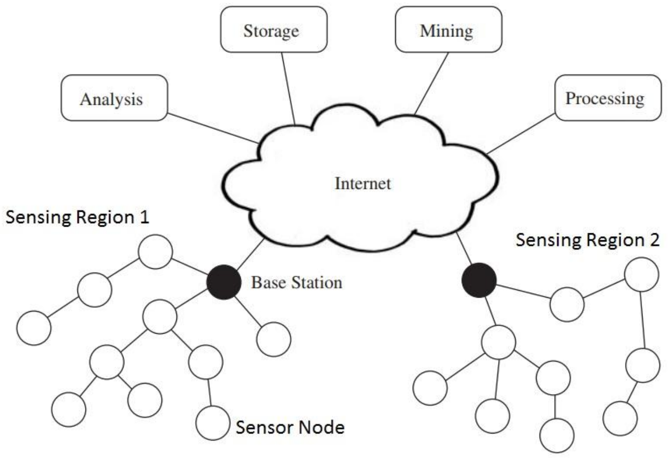
5. IoT API Security

This technology enables us to authenticate and authorize data movement between IoT devices, back-end systems, and applications using documented REST-based APIs. API security protects the integrity of data transiting between edge devices and back-end systems, and applications using documented rested APIs as well as detecting potential threats and attacks against APIs. Sample vendors: Akana, Apigee/Google, Axway, CA Technologies, Mashery/TIBCO, MuleSoft, and more

2.5 IOT(INTERNET OF THINGS) ENABLING TECHNOLOGIES ARE

1. Wireless Sensor Network
2. Cloud Computing
3. Big Data Analytics
4. Communications Protocols
5. Embedded System

1.WIRELESS SENSOR NETWORK(WSN) :



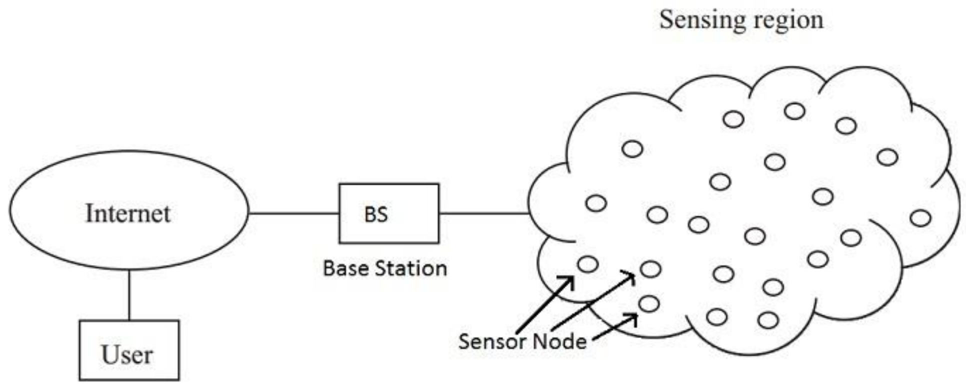
A **WSN** comprises distributed devices with sensors which are used to monitor the environmental and physical conditions. A **wireless sensor network** consists of end nodes, routers and coordinators. End nodes have several sensors attached to them where the data is passed to a coordinator with the help of routers. The coordinator also acts as the gateway that connects WSN to the internet.
Example –

- Weather monitoring system
- Indoor air quality monitoring system

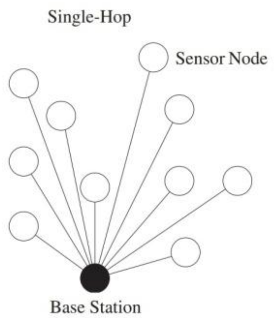
- Soil moisture monitoring system
- Surveillance system
- Health monitoring system

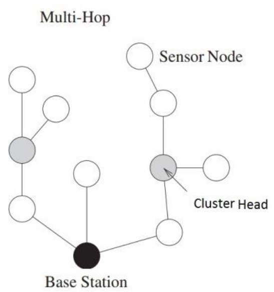
NETWORK ARCHITECTURE

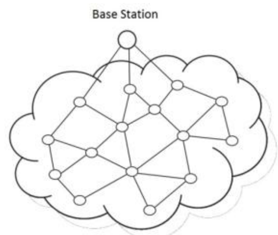
When a large number of sensor nodes are deployed in a large area to co-operatively monitor a physical environment, the networking of these sensor node is equally important. A sensor node in a WSN not only communicates with other sensor nodes but also with a Base Station (BS) using wireless communication.

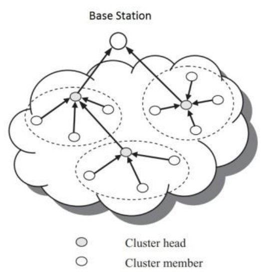


The base station sends commands to the sensor nodes and the sensor node perform the task by collaborating with each other. After collecting the necessary data, the sensor nodes send the data back to the base station.
A base station also acts as a gateway to other networks through the internet. After receiving the data from the sensor nodes, a base station performs simple data processing and sends the updated information to the user using internet.
If each sensor node is connected to the base station, it is known as Single-hop network architecture. Although long distance transmission is possible, the energy consumption for communication will be significantly higher than data collection and computation.









Hence, Multi-hop network architecture is usually used. Instead of one single link between the sensor node and the base station, the data is transmitted through one or more intermediate node.

This can be implemented in two ways. Flat network architecture and Hierarchical network architecture. In flat architecture, the base station sends commands to all the sensor nodes but the sensor node with matching query will respond using its peer nodes via a multi-hop path.

In hierarchical architecture, a group of sensor nodes are formed as a cluster and the sensor nodes transmit data to corresponding cluster heads. The cluster heads can then relay the data to the base station.

BENEFITS OR ADVANTAGES OF WSN

Following are the benefits or **advantages of WSN**:
➨It is scalable and hence can accommodate any new nodes or devices at any time.
➨It is flexible and hence open to physical partitions.
➨All the WSN nodes can be accessed through centralized montoring system.

➨As it is wireless in nature, it does not require wires or cables. Refer difference between wired network vs wireless network.
➨WSNs can be applied on large scale and in various domains such as mines, healthcare, surveillance, agriculture etc.
➨It uses different security algorithms as per underlying wireless technologies and hence provide reliable network for consumers or users.

DRAWBACKS OR DISADVANTAGES OF WSN

Following are the drawbacks or **disadvantages of WSN**:
➨As it is wireless in nature, it is prone to hacking by hackers.
➨It can not be used for high speed communication as it is designed for low speed applications.
➨It is expensive to build such network and hence can not be affordable by all.
➨There are various challenges to be considered in WSN such as energy efficiency, limited bandwidth, node costs, deployment model, Software/hardware design constraints and so on.
➨In star topology based WSN, failure of central node leads to whole network shutdown.

COMPONENTS OF WSN:

1. **Sensors:**
   Sensors in WSN are used to capture the environmental variables and which is used for data acquisition. Sensor signals are converted into electrical signals.Sensors may be Arial, terrestrial and underwater.
2. **Radio Nodes:**

   It is used to receive the data produced by the Sensors and sends it to the WLAN access point. It consists of a microcontroller, transceiver, external memory, and power source.
3. **WLAN Access Point:**

   It receives the data which is sent by the Radio nodes wirelessly, generally through the internet.
4. **Evaluation Software:**

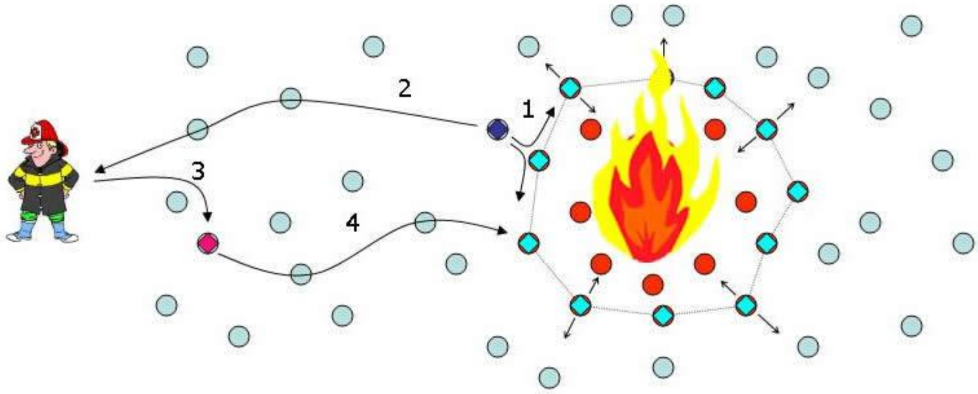
   The data received by the WLAN Access Point is processed by a software called as Evaluation Software for presenting the report to the users for further processing of the data which can be used for processing, analysis, storage, and mining of the data.

APPLICATIONS OF WSN:

There are numerous applications of WSNs in industrial automation, traffic monitoring and control, medical device monitoring and in many other areas. Some of applications are discussed below:

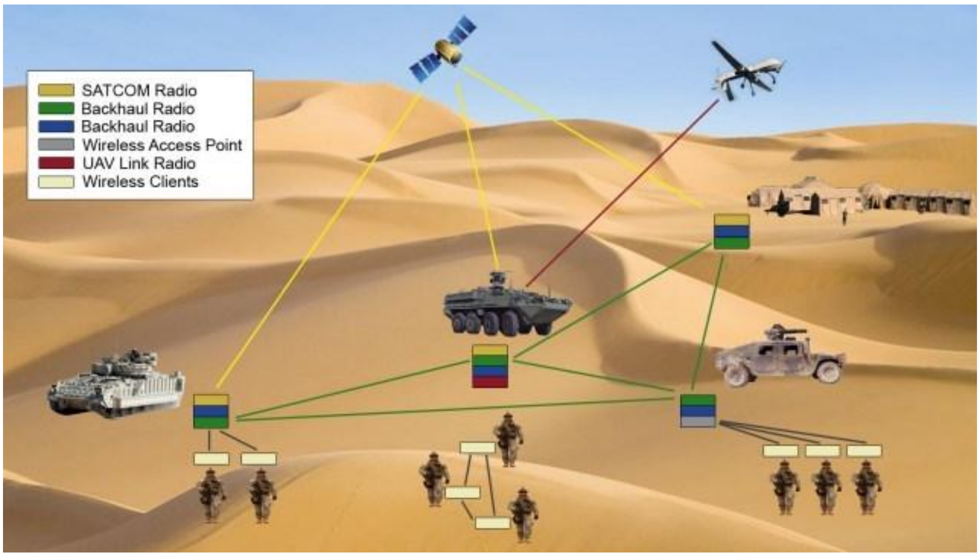
1. DISASTER RELIEF OPERATION

If an area is reported to have been stricken from some sort of calamity such as wildfire, then drop the sensor nodes on the fire from an aircraft. Monitor the data of each node and construct a temperature map to devise proper ways and techniques to overcome the fire.



2. MILITARY APPLICATIONS

As the WSNs can be deployed rapidly and are self organized therefore they are very useful in military operations for sensing and monitoring friendly or hostile motions. The battlefield surveillance can be done through the sensor nodes to keep a check on everything in case more equipment, forces or ammunitions are needed in the battlefield. The chemical, nuclear and biological attacks can also be detected through the sensor nodes.



An **example** of this is the 'sniper detection system' which can detect the incoming fire through acoustic sensors and the position of the shooter can also be estimated by processing the detected audio from the microphone.

3. ENVIRONMENTAL APPLICATIONS

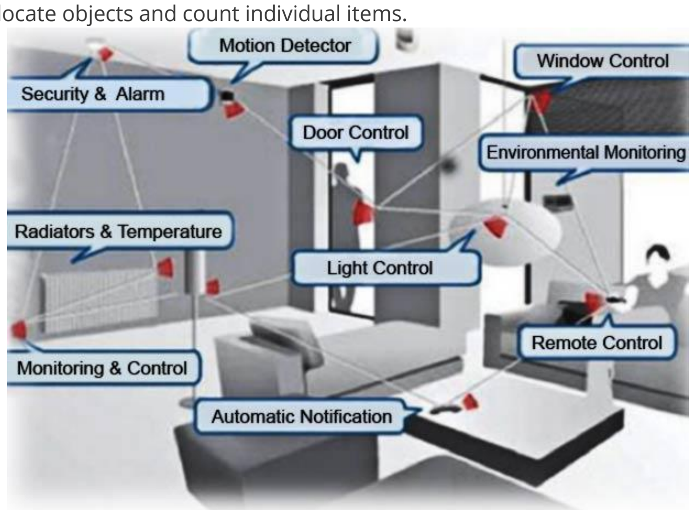
These sensor networks have a huge number of applications in the environment. They can be used to track movement of animals, birds and record them. Monitoring of earth, soil, atmosphere context, irrigation and precision agriculture can be done through these sensors. They can also used for the detection of fire, flood, earthquakes, and chemical/biological outbreak etc.

A common **example** is of 'Zebra Net'. The purpose of this system is to track and monitor the movements and interactions of zebras within themselves and with other species also.

4. MEDICAL APPLICATIONS

In health applications, the integrated monitoring of a patient can be done by using WSNs. The internal processes and movements of animals can be monitored. Diagnostics can be done. They also help in keeping a check on drug administration in hospitals and in monitoring patients as well as doctors.

An **example** of this is 'artificial retina' which helps the patient in detecting the presence of light and the movement of objects. They can also locate objects and count individual items.



5. HOME APPLICATIONS

As the technology is advancing, it is also making its way in our household appliances for their smooth running and satisfactory performance. These sensors can be found in refrigerators, microwave ovens, vacuum cleaners, security systems and also in water monitoring systems. The user can control devices locally as well as remotely with the help of the WSNs.

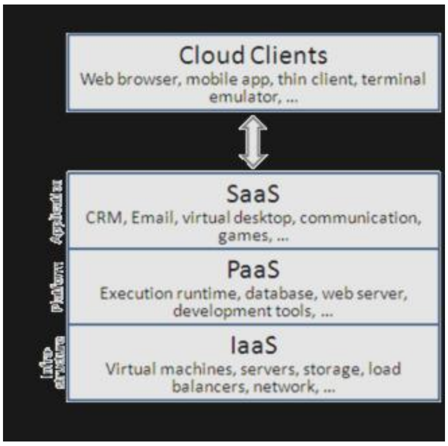
2. CLOUD COMPUTING :

It provides us the means by which we can access applications as utilities over the internet. Cloud means something which is present in remote locations.
With Cloud computing, users can access any resources from anywhere like databases, webservers, storage, any device, and any software over the internet.

THE CHARACTERISTICS OF CLOUD COMPUTING ARE:

- **On demand:** The resources in the cloud are available based on the traffic. If the incoming traffic increases, the cloud resources scale up accordingly and when the traffic decreases, the cloud resources scale down accordingly.
- **Autonomic:** The resource provisioning in the cloud happens with very less to no human intervention. The resources scale up and scale down automatically.
- **Scalable:** The cloud resources scale up and scale down based on the demand or traffic. This property of cloud is also known as elasticity.
- **Pay-per-use:** On opposite to traditional billing, the cloud resources are billed on pay-per-use basis. You have to pay only for the resources and time for which you are using those resources.
- **everywhere:** You can access the cloud resources from anywhere in the world from any device. All that is needed is Internet. Using Internet you can access your files, databases and other resources in the cloud from anywhere.

PROVIDES DIFFERENT SERVICES, SUCH AS –



- **IaaS (**Infrastructure as a service**)**

  Infrastructure as a service provides online services such as physical machines, virtual machines, servers, networking, storage and data center space on a pay per use basis. Major IaaS providers are Google Compute Engine, Amazon Web Services and Microsoft Azure etc.
  Ex : Web Hosting, Virtual Machine etc.

- **PaaS (**Platform as a service**)**

  Provides a cloud-based environment with a very thing required to support the complete life cycle of building and delivering West web based (cloud) applications – without the cost and complexity of buying and managing underlying hardware, software provisioning and hosting. Computing platforms such as hardware, operating systems and libraries etc. Basically, it provides a platform to develop applications.
  Ex : App Cloud, Google app engine

- **SaaS (**Software as a service**)**

  It is a way of delivering applications over the internet as a service. Instead of installing and maintaining software, you simply access it via the internet, freeing yourself from complex software and hardware management.SaaS Applications are sometimes called web-based software on demand software or hosted software.SaaS applications run on a SaaS provider's service and they manage security availability and performance.
  Ex : Google Docs, Gmail, office etc.

3. BIG DATA ANALYTICS :

It refers to the method of studying massive volumes of data or big data. Collection of data whose volume, velocity or variety is simply too massive and tough to store, control, process and examine the data using traditional databases. Big data is gathered from a variety of sources including social network videos, digital images, sensors and sales transaction records.

Several steps involved in analyzing big data –

1. Data cleaning
2. Managing
3. Processing
4. Visualization

Examples –

- Sensor data generated by IoT system such as weather monitoring stations.
- Machine sensor data collected from sensors embedded in industrial and energy systems for monitoring their health and detecting Failures.
- Health and fitness data generated by IoT devices such as wearable fitness bands
- Data generated by ioT systems for location and tracking of vehicles
- Data generated by retail inventory monitoring systems

CHARACTERISTICS

Big data can be described by the following characteristics:

- **Volume –** The quantity of generated and stored data. The size of the data determines the value and potential insight, and whether it can be considered big data or not.
- **Variety –** The type and nature of the data. This helps people who analyze it to effectively use the resulting insight. Big data draws from text, images, audio, video; plus it completes missing pieces through data fusion.
- **Velocity(speed) –** In this context, the speed at which the data is generated and processed to meet the demands and challenges that lie in the path of growth and development. Big data is often available in real-time. Compared to small data, big data are produced more continually. Two kinds of velocity related to Big Data are the frequency of generation and the frequency of handling, recording, and publishing.
- **Veracity(reality) –** It is the extended definition for big data, which refers to the data quality and the data value. The data quality of captured data can vary greatly, affecting the accurate analysis.

BIG DATA ANALYTICS BENEFITS

- Quickly analyzing large amounts of data from different sources, in many different formats and types.
- Rapidly making better-informed decisions for effective strategizing, which can benefit and improve the supply chain, operations and other areas of strategic decision-making.
- Cost savings, which can result from new business process efficiencies and optimizations.
- A better understanding of customer needs, behavior and sentiment, which can lead to better marketing approach, as well as provide information for product development.
- Improved, better informed risk management strategies that draw from large sample sizes of data.

HOW BIG DATA ANALYTICS WORKS

1. Collect Data

- Data collection looks different for every organization. With today's technology, organizations can gather both structured and unstructured data from a variety of sources — from cloud storage to mobile applications to in-store IoT sensors and beyond. Some data will be stored in data warehouses where business intelligence tools and solutions can access it easily. Raw or unstructured data that is too diverse or complex for a warehouse may be assigned metadata and stored in a data lake.

2. Process Data

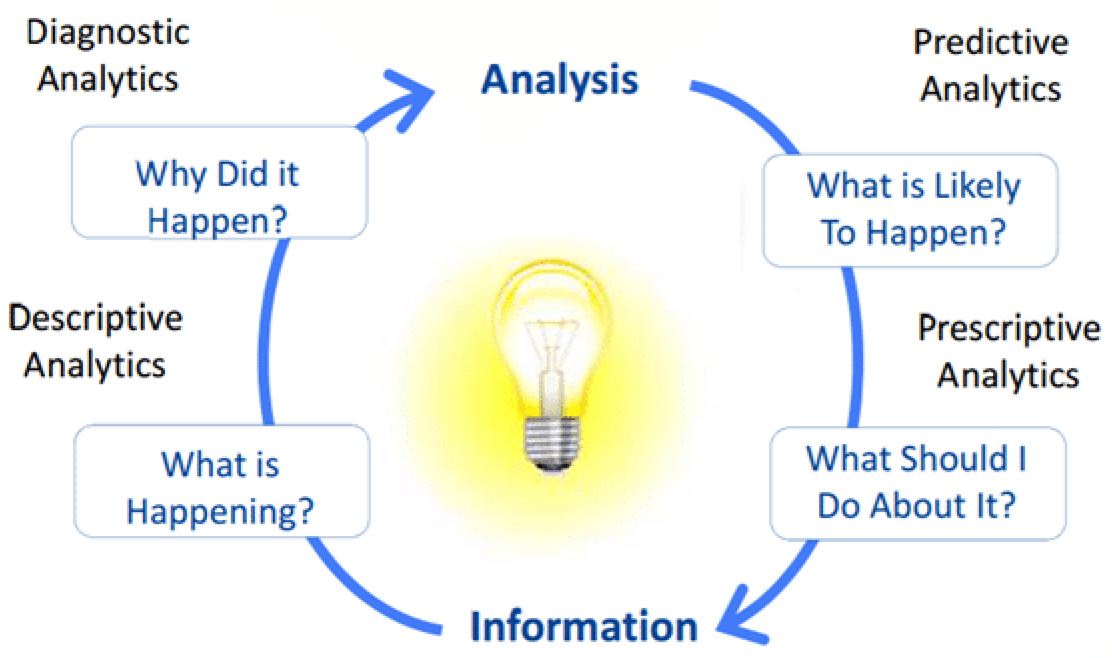
- Once data is collected and stored, it must be organized properly to get accurate results on analytical queries, especially when it's large and unstructured. Available data is growing exponentially, making data processing a challenge for organizations. One processing option is **batch processing**, which looks at large data blocks over time. Batch processing is useful when there is a longer turnaround time between collecting and analyzing data. **Stream processing** looks at small batches of data at once, shortening the delay time between collection and analysis for quicker decision-making. Stream processing is more complex and often more expensive.

3. Clean Data

- Data big or small requires scrubbing to improve data quality and get stronger results; all data must be formatted correctly, and any duplicative or irrelevant data must be eliminated or accounted for. Dirty data can obscure and mislead, creating flawed insights.

4. Analyze Data

- Getting big data into a usable state takes time. Once it's ready, advanced analytics processes can turn big data into big insights. Some of these big data analysis methods include:
- **Data mining** sorts through large datasets to identify patterns and relationships by identifying anomalies and creating data clusters.
- **Predictive analytics** uses an organization's historical data to make predictions about the future, identifying upcoming risks and opportunities.
- **Deep learning** imitates human learning patterns by using artificial intelligence and machine learning to layer algorithms and find patterns in the most complex and abstract data.
- Types of Big Data Analytics



- Big data analytics is categorized into four subcategories that are:
    - Descriptive Analytics
    - Diagnostic Analytics
    - Predictive Analytics
    - Prescriptive Analytics

1. Descriptive Analytics

- Descriptive Analytics is considered a useful technique for uncovering patterns within a certain segment of customers. It simplifies the data and summarizes past data into a readable form.
- Descriptive analytics provide what has occurred in the past and with the trends to dig into for more detail. This helps in creating reports like a company's revenue, profits, sales, and so on.
- Examples of descriptive analytics include summary statistics, clustering, and association rules used in market basket analysis.
- An example of the use of descriptive analytics is the Chemical Company. The company utilized its past data to increase its facility utilization across its offices and labs.

2.Diagnostic Analytics

- Diagnostic Analytics, as the name suggests, gives a diagnosis to a problem. It gives a detailed and in-depth approaching into the root cause of a problem.
- Data scientists turn to this analytics want for the reason behind a particular happening. Techniques like drill-down, data mining, and data recovery, mix reason analysis, and customer health score analysis are all examples of diagnostic analytics.
- In business terms, diagnostic analytics is useful when you are researching the reasons leading mix indicators and usage trends among your most loyal customers.

3.Predictive Analytics(Descriptive+diagnostic)

- Predictive Analytics, as can be recognized from the name itself, is concerned with predicting future incidents. These future incidents can be market trends, consumer trends, and many such market-related events.

- This type of analytics makes use of historical and present data to predict future events. This is the most commonly used form of analytics among businesses.

- Predictive analytics doesn't only work for the service providers but also for the consumers. It keeps track of our past activities and based on them, predicts what we may do next.

- Examples of Predictive analytics include next best offers, churn(mix) risk, and renewal risk analysis.

4. Prescriptive Analytics

- Prescriptive analytics is the most valuable yet underused form of analytics. It is the next step in predictive analytics. The prescriptive analysis explores several possible actions and suggests actions depending on the results of descriptive and predictive analytics of a given dataset.

- Prescriptive analytics is a combination of data and various business rules. The data of prescriptive analytics can be both internal (organizational inputs) and external (social media insights).

- Prescriptive analytics allows businesses to determine the best possible solution to a problem. When combined with predictive analytics, it adds the benefit of manipulating a future occurrence like ease future risk.

- Examples of prescriptive analytics for customer maintenance is the next best action and next best offer analysis.

4. COMMUNICATIONS PROTOCOLS :

They are the backbone of IoT systems and enable network connectivity and linking to applications. Communication protocols allow devices to exchange data over the network. Multiple protocols often describe different aspects of a single communication. A group of protocols designed to work together is known as a protocol suite; when implemented in software they are a protocol stack. They are used in

1. Data encoding
2. Addressing schemes

EMBEDDED SYSTEMS :

It is a combination of hardware and software used to perform special tasks. It includes microcontroller and microprocessor memory, networking units (Ethernet Wi-Fi adapters), input output units (display keyword etc.) and storage devices (flash memory). It collects the data and sends it to the internet. Embedded systems used in Examples –

- Digital camera
- DVD player, music player
- Industrial robots
- Wireless Routers etc.

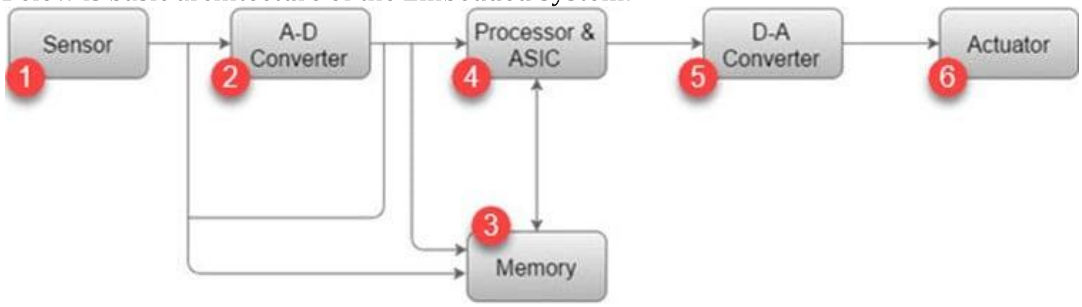
An embedded system has three components.

1. Hardware
2. Software

3. Real Time Operating system (RTOS) that supervises the application software and provide mechanism to let the processor run a process as per schedule by following a plan to control the latencies

ARCHITECTURE OF THE EMBEDDED SYSTEM

- Below is basic architecture of the Embedded System:



Architecture of the Embedded System

1) Sensor:

Sensor helps you to measures the physical quantity and converts it to an electrical signal. It also stores the measured quantity to the memory. This signal can be ready by an observer or by any electronic instrument such as A2D converter.

2) A-D Converter:

A-D converter (analog-to-digital converter) allows you to convert an analog signal sent by the sensor into a digital signal.

3) Memory:

Memory is used to store information. Embedded System majorly contains two memory cells 1) Volatile 2) Non volatile memory.

4) Processor & ASICs(Application-specific integrated circuit):

This component processes the data to measure the output and store it to the memory.

5) D-A Converter:

D-A converter (A digital-to-analog converter) helps you to convert the digital data fed by the processor to analog data.

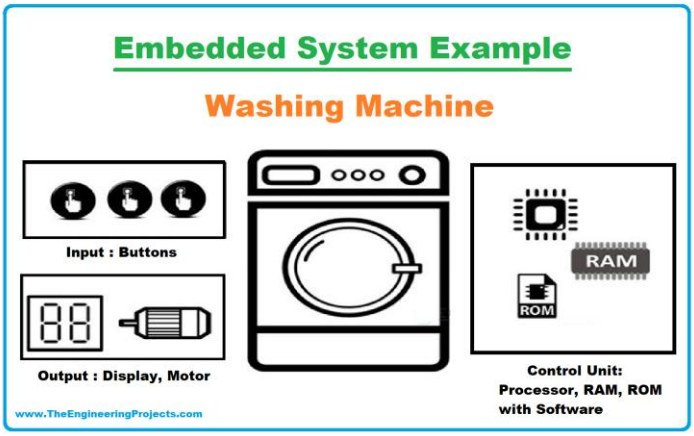
6) Actuator:

An actuator allows you to compare the output given by the D-A converter to the actual output stored in it and stores the approved output in the memory.

THE CHARACTERISTICS OF AN EMBEDDED SYSTEM ARE:

1. Specific(Single functional)

- Embedded systems can either be domain-specific or task-specific.
- An embedded system is designated to perform the dedicated task only, it cannot be made to perform several automated functions from different inputs.
- For example, you can only wash clothes in a washing machine, it cannot cook food so this is a task-specific embedded system.
- Meanwhile, domain-specific embedded systems fall under certain domains or categories such as a mobile embedded system is a domain-specific embedded system.



2. Strict Design Parameters(tightly constrained)

- The design metrics are pre-defined for every system but they are configurable to a great extent, or we can say there is room for additions and extensions in systems other than embedded systems.
- But there is very little room for extensions and additions in an embedded because we have to fix (cost, size, performance, power) everything on a single chip, that can perform its designated function independently, as we have already discussed the components of an embedded system you can understand this very well by now!

3. Efficiency(real time/timely)

- An embedded system must be efficient enough to react and respond to the real-time environment.
- In certain real-time embedded systems, a system has to react according to the real-time situation and adjust accordingly such as the air conditioner works on the same principle.
- The speed control of a car also works in a real-time response manner by reacting to the real-time situation hence managing the speed and brakes.

4. Microprocessor Or Microcontroller Based

- The Embedded system must have a microcontroller or microprocessor.

Features of A Microcontroller: A microcontroller is different from a microprocessor, and a microcontroller has the following features being listed below;

- Most of the time a microcontroller is used according to the system requirements, which comes in different sizes such as 6 bit, 8 bit, 16 bit and 32 bit.
- The microcontroller is composed of an external processor and internal memory along with I/O components.
- It consumes less power because everything is present internally on the chip. A lot of microcontrollers have a power-saving mode as well.
- It is easier to write a program on the microcontroller.

5. Exclusive Memory

- Embedded systems don't have a secondary memory, their memory is present in embedded software in the form of ROM.

6. Multi-Rate Operational System

- Some of the large-scale embedded systems are multi-rate operational systems i.e a large number of embedded systems are performing their dedicated functions independently to run a system.
- All the multi-rate operational, embedded systems are well articulated to work synchronously with each other.
- For example, many embedded systems work independently to keep a car on road.

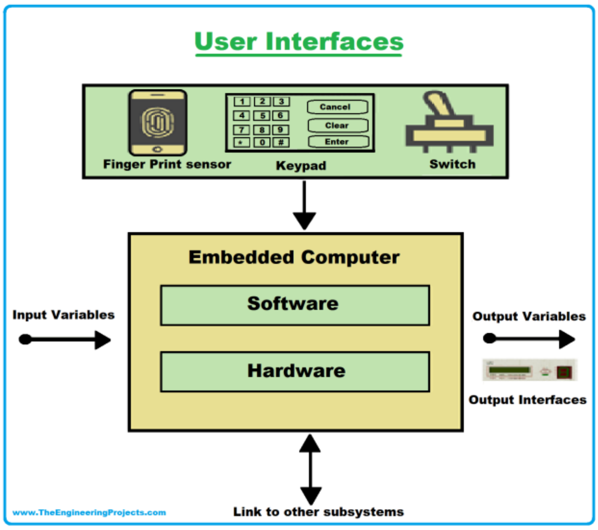
7. Compact Design

- An embedded system is designed to compact and lightweight as everything is to be placed on a single chip to perform a task including the microcontroller, timer, I/O parts, and the embedded software as well.

8. Minimal Power waste

- Embedded systems are designed in such a way to waste power at its minimal.
- The goal is to conserve power and prevent overheating of the system by adding in heat sinks and cooling fans, and sometimes a larger battery is used to run the system.

9. Minimal User Interface



- Have a brief look at your air conditioner, your oven, or your washing machine, you might have noticed one thing in common, a minimal user interface.
- Embedded systems are designed with minimal user interface because users have almost nothing to do by themselves, you only have to provide the input or we can say instructions, the system is already fully automated to perform the designated task accordingly!

10. Safety Factor

- A safety analysis is always necessarily carried out for the embedded systems to ensure the safety of the operator and the environment in case of any material damage or dangerous production from the system.

12. Cost-Effective

- An embedded system is designed in such a way to make it cost-effective, overall the circuit is small since everything is present on a single chip.
- A compact design and designated functionality make an embedded system less costly and high speed.